

Physics-Informed Neural Digital Twin (PI-NDT)

Real-Time 3D Battery State Estimation & Autonomous Fast-Charging Safety

Project Scope: 100% Software
(Microcontroller-Ready)

Domain: Deep Learning, Physics-Informed
ML, Autonomous Agents

Target Impact: Lithium-Ion Safety & Zero-
Fire EV Fast Charging

1. Executive Summary

PROJECT VISION

This project develops an AI-powered **Physics-Informed Neural Digital Twin** that acts like a real-time "X-ray" for lithium-ion batteries in electric vehicles, consumer tech, and energy storage. Standard sensors can only measure the exterior metal casing of a battery, leaving dangerous, microscopic internal core hotspots completely invisible. By feeding non-invasive surface measurements into a neural network whose loss function is mathematically governed by thermodynamic heat-conduction and electrochemical partial differential equations (PDEs), our software reconstructs the invisible 3D internal temperature and degradation field in **under 2 milliseconds**—over 1,000× faster than traditional engineering simulations. An integrated multi-agent supervisory loop monitors these internal thermal maps in real time, dynamically optimizing charging current to achieve maximum fast-charging speeds while eliminating thermal runaway and fire hazards.

2. The Real-World Industrial Crisis

2.1 The Sensor Blind Spot & Thermal Lag

Lithium-ion cells are tightly sealed chemical units. Placing physical thermocouples or sensors inside the core of a cell is physically impossible in production because metallic probes puncture separators and trigger violent short-circuits. Consequently, automotive and phone manufacturers place sensors **only on the outside casing**.

However, battery materials are poor thermal conductors. When high current flows, localized Joule heating and internal resistance heat the inner jelly-roll core to **75°C+**, while the exterior metal casing still registers a seemingly safe **35°C**. By the time this internal heat reaches exterior sensors, the cell has already begun catastrophic, irreversible thermal runaway.

Case in Point — The \$2 Billion GM Recall: In 2021, General Motors was forced to recall over 140,000 Chevrolet Bolt EVs at an estimated cost of \$2 Billion due to battery cells catching fire during charging. Standard Battery Management Systems (BMS) were completely blind to internal microscopic defects because surface temperature telemetry stayed normal right up until ignition.

2.2 The Fast-Charging Throttling Dilemma

Every modern EV driver expects 15-minute ultra-fast charging at 350 kW stations. However, because battery management computers cannot see inside the core, car manufacturers are forced to use static, defensive

safety cutoffs. Charging stations prematurely throttle power down to 50 kW after just 8 minutes, tripling charging stop times to 45 minutes purely out of fear of the unknown.

3. Machine Learning Methodology

A neural network is the foundation of **Deep Learning**, which is a core branch of Machine Learning. Unlike classical numerical solvers (COMSOL or ANSYS) that take 10 minutes to compute differential equations and crash on noisy real-world data, our machine learning pipeline solves the problem at millisecond speeds using three integrated ML pillars:

Machine Learning Area	Algorithm / Architecture	Core Function in System
1. Deep Learning (Physics-Informed)	Physics-Informed Neural Network (PINN) / Fourier Neural Operator (FNO)	Inverts sparse exterior surface inputs into full continuous 3D internal thermal and lithium-ion concentration fields in <2 ms.
2. Classical Unsupervised ML	Isolation Forest / One-Class Support Vector Machine (SVM)	Continuously scores predicted internal state trajectories against baseline healthy cells, detecting microscopic short-circuits and dendrites.
3. Reinforcement Learning (RL)	Proximal Policy Optimization (PPO) / Deep Q-Networks	Autonomous control agent that dynamically modulates charger amperage to maximize charging rate while guaranteeing safe internal temperatures.

The Physics-Informed Loss Function (Learning Without Internal Sensors)

Standard supervised ML requires millions of labeled internal measurements, which cannot physically be acquired. Physics-Informed ML bypasses this limitation by using the **laws of thermodynamics as the teacher**:

$$\text{Loss_Total} = \text{Loss_Surface_Data} + \lambda \cdot \text{Loss_Physics_Violation}$$

Where the physics loss penalizes violations of Fourier's Law of Heat Conduction:

$$\rho \cdot c_p \cdot (\partial T / \partial t) - \nabla \cdot (k \cdot \nabla T) = Q_{\text{generation}}(\text{Current}, \text{Voltage})$$

If the neural network guesses an internal core temperature that violates the conservation of energy, the backpropagation algorithm penalizes the weights and corrects the model. This allows the neural network to deduce exact internal temperatures without ever having internal sensors installed.

4. Autonomous Multi-Agent Software Architecture

The system is organized into four specialized software agents operating in a continuous real-time control loop:

Agent 1: Digital Twin Observer Agent

Physics-Informed Neural Network (PINN) Engine

Ingests live telemetry streams (Current I , Voltage V , Surface Temp T_{surf}). Reconstructs the real-time 3D internal temperature and chemical stress gradients in 1.5 milliseconds.

Agent 2: Diagnostic Critic Agent

Unsupervised Anomaly Detector (Isolation Forest & Drift Metric)

Monitors predicted core thermal gradients. If one cell heats $2\times$ faster than surrounding cells under identical current, it flags an internal micro-short circuit or localized dendrite growth.

Agent 3: Supervisory Action Agent

Reinforcement Learning / Safe Model Predictive Control (MPC)

Controls the charging current. Balances maximum power delivery against a strict hard boundary: keeping the predicted internal core temperature strictly below 60°C .

Agent 4: Explainer & Reporting Agent

Automated Telemetry Summarizer & Diagnostics Logger

Synthesizes real-time decisions into structured diagnostic logs (e.g., "Cycle #240: Throttled current by 12% at min 9 due to core thermal gradient exceeding $15^{\circ}\text{C}/\text{cm}$; preserved estimated lifespan by 18%").

5. Implementation Strategy: Software Today, Hardware Tomorrow

Phase 1: Pure Software Development (Immediate Scope)

- **Zero Hardware Requirement:** Executes entirely on a standard laptop or free Google Colab environment.
- **Synthetic Simulation Engine:** Uses PyBaMM (Python Battery Mathematical Modelling) to generate multi-rate charging cycles (1C, 2C, 3C fast-charging).
- **Real-World Benchmark Datasets:** NASA Ames Prognostics Center of Excellence (PCoE) Battery Aging Dataset and Oxford Battery Degradation Dataset.
- **Demonstration UI:** An interactive Streamlit and Plotly web dashboard displaying animated 3D heat cross-sections of battery cells in real time with dynamic safety alerts.

Phase 2: Edge Hardware Extension (Future Development)

- **Quantization:** Export the trained PyTorch model to ONNX Runtime or TensorFlow Lite for Microcontrollers (<150 KB footprint).
- **Edge Deployment:** Flash the model onto an inexpensive \$6 **ESP32** or **Raspberry Pi Pico** microcontroller.
- **Physical Setup:** Connect 2–3 digital thermistors (DS18B20) and an INA219 current/voltage monitor to an external 18650 lithium-ion cell holder.
- **Outcome:** A commercializable, physical next-generation Smart Battery Management System (BMS).

6. 4-Week Milestone Execution Roadmap

Timeline	Focus Area	Action Items	Milestone Output
Week 1	Physics Simulation Sandbox	Install pybamm. Simulate 18650 cylindrical cells under 1C, 2C, and 3C fast-charge protocols. Log surface and core thermal trajectories.	Baseline synthetic dataset & verified physics simulation pipeline.
Week 2	PINN Model Development	Build a forward heat-equation PINN in PyTorch/DeepXDE. Implement custom loss function penalizing thermodynamic PDE residuals.	Converged PINN solving forward thermal diffusion without divergence.
Week 3	Inverse State Estimation	Train model strictly on surface boundary data to estimate invisible core temperatures. Benchmark against NASA open battery aging cycles.	Validated inverse state model (<1.5°C RMSE accuracy vs. ground truth).
Week 4	Agents & Interactive UI	Integrate the Anomaly Critic and Supervisory Action Agent. Build an interactive Streamlit 3D visualization dashboard.	Polished GitHub repository, live web demo, and project documentation.

7. Target Technical Specifications

Metric	Target Value	Industry Baseline Comparison
Inference Latency	< 2.0 milliseconds	Finite Element Simulators (COMSOL): 30 to 300 seconds
Core Temperature Estimation Error	< 1.5°C RMSE	Linear Equivalent Circuit Models (ECM): 5.0°C to 8.0°C Error
Fast-Charging Time Improvement	25% – 35% Time Saved	Conservative static throttling curves
Model Size (Quantized)	< 200 KB	Easily fits inside low-cost microcontrollers (\$5 ESP32)